A Overall workflow

1.1 The Problem faced in Cardiovascular diagnosis

Cardiovascular disease is leading to cause of death in worldwide. It is affecting young people in their 20s,30s and early 40s.Driven through by poor diet,sedentary life style and stress and smoking in India 25% of heart attacks occur in people under 40.

Coronary artery disease, heart valve disease, thoracic aorta disease, heart rhythm disease and cardio-oncology are the most common heart pathologies

The current modern system uses algorithms like “XGBoost” builds thousands of simple “Decision Trees"Each tree fix problems or mistakes made by previous trees.
Are system uses (Sparse FCM + DBN + Taylor-BSA) this mathematically guarantees it finds the absolute error rate.

This both algorithm has there “pros” and “cons”.

1.2 System architecture

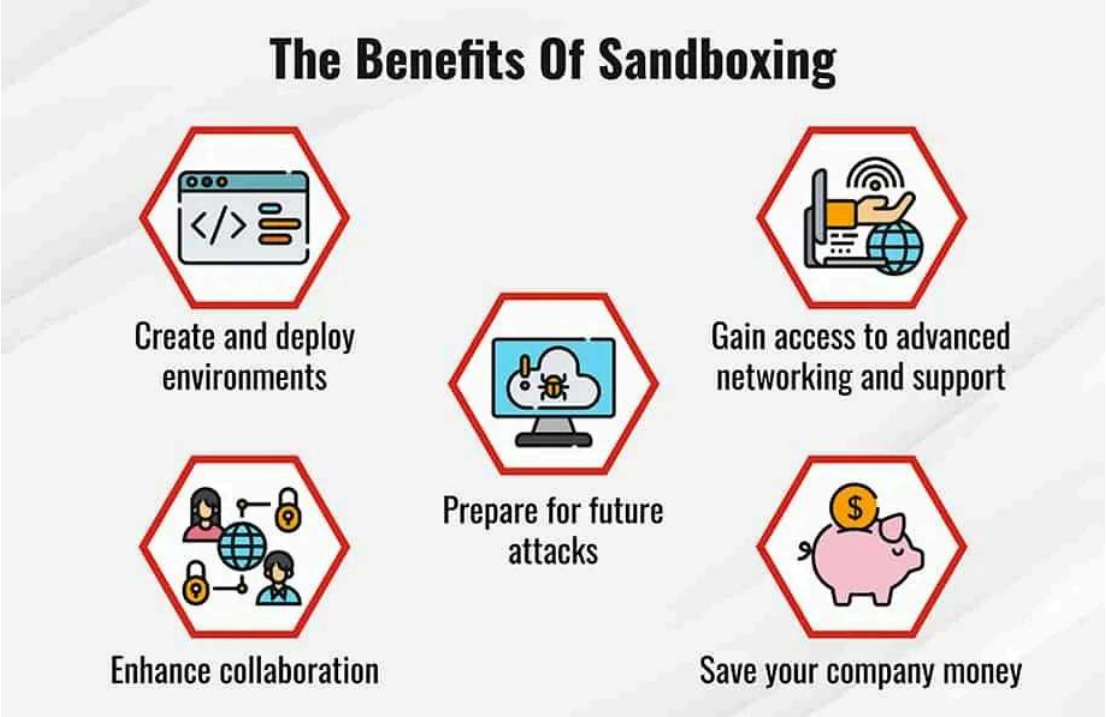
1.Pakages and necessary tools need to download

The system or the backend will be based on 70-80% Python language and combination of C/C++ and SQL(Database) and Web (Fronted)

Using combination of built in and third party libraries in python library like:
1.pytourch ai
2.
3.
..

But Before Downloading all mentioned packages above we First have to make a local copy of python interpreter i.e a creation of “Sandbox” to insure

that the code or the project could work on other person's laptop or PC.



To ensure users dont have to download packages manually we can make a “Shell script” that will download all the necessary packages needed.(Resulting into automation)

2.Dataset for training the Ai model.

The Dataset is chosen is from **UCI Machine Learning Repository (Cleveland Clinic Foundation)** a globally recognized dataset for ai researchers folder name “processed.cleveland.data”
Its a dataset from 1988 and the format of the data is : Multivariable, Tabular and data type is a mixture of float with integer values.
The dimensions of the data are Total Rows: 303 | Train: 242 | Test: 61 | Columns: 14.

As Data in number is too big from the file we write a script that takes a massive number ((like 233.0 cholesterol) to small decimal values between -1 and 1. So the Ai does not get overwhelmed

3.AI Model Architecture & Algorithms

In this there will be folder with three Algorithms namely
1.Sparse FCM
2.Deep Belief Network
3.Tylor-BSA

The above algorithms play a fundamental role in this project as they are the heart of the whole project.

3.1 Sparse FCM(The filter)

So a medical dataset often contains noise or unwanted or irrelevant data (noise) to confuse an AI . To solve this we implement this algorithm as a pre-processing filter. Instead of manually guessing which symptoms matter this algorithm mathematically calculates which symptoms matter for each clinical feature. It assigns high weight to the critical symptoms and filters out the nose. Ensuring the Ai model learns only from the important data.

3.2 Deep belief network (The classifier)

The core diagnostic engine is Deep belief network (DBN) unlike basic old tradition algorithms like CNN .The DBN is a probabilistic generative model that is well suited for medical use. It is built by stacking up multiple Restricted Boltzmann's Machines (RBMs).It has two layers “Visible layer” and “Hidden layer”.Is is called restricted because connection exists only between layers not within them.The model learns in stages:The lower layer process the medical metrics(like age or resing heart rate) while the higher level combine these metrics to uncover some hidden complex patterns that a human doctor might miss.

3.3 Taylor-BSA (Optimizer)

This algorithm fine tunes the DBN it behaviour is like a bird(Swarm intelligence-mimicking the communication and foraging behavior of flock of birds) its features are:

1.Foraging (Exploration: Searches for food based on experience)
2.Vigilance (Safety: Towards center avoids collusion)
3.Flight (Randomization: Random location)

This Navigates through complex DBN landscape (More efficiently). It does this mathematically and explores multiple solutions simultaneously,ensuring the network finds the absolute lowest Cross-Entropy loss and the highest possible accuracy.

4.Database architecture

4.1 Database choice

The system will utilize a relational database management system(RDBMS) such as Mysql to manage all the data.because DBN uses strictly 13 specific numerical input functions , medical data is highly structured and tabular we use relational SQL because it enforces ACID compliance to the data that will be entered and hence it maintains absolute data integrity.

4.2 Table structure

The database will be normalized into two primary, interconnected tables to separate the raw medical inputs from the AI's calculated outputs.

Table1:

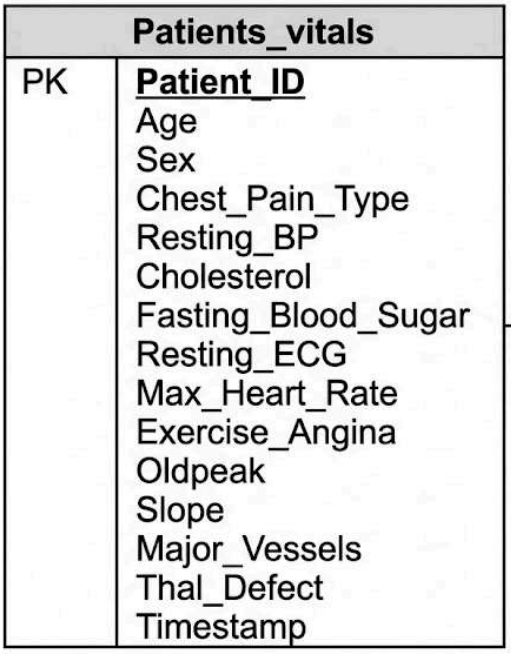
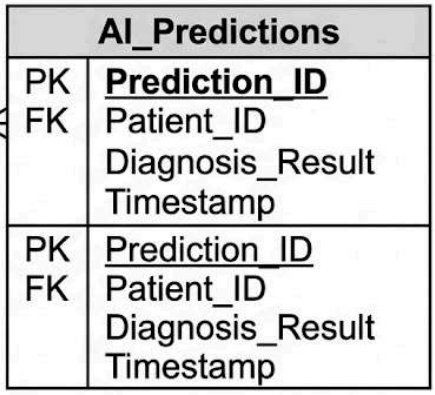


Table2:



4.3 System Data flow

To ensure there is a smooth flow of data and reliable processing ,the data follows a strict ,one - way execution from user interface to ai model and back

1.Data capture [Fronted]

The attending doctor inputs the 13 clinical metrics into the secure web UI and click the analysis button

2. Database insertion [Backend to sql]

The web backend receives the data and immediately inserts a new row
Into the SQL table (Patients_value) to ensure the raw medical record is safely stored before any processing begins

3. Ai Execution [Python AI Engine]

The backend script fetches this new row, normalizes the numbers, and feeds the 13 variables directly into the frozen “heart_disease_model” file.

4. Result logging [AI to SQL]

The Deep Belief Network calculates the diagnosis. Before showing the doctor, the backend saves this result (0 or 1) into the “AI_prediction” table, permanently linking it to the patient's ID.

5. Physician Display (Doctor) [Fronted]

The final diagnosis is pushed back to the website dashboard, presenting a clear, color-coded "Healthy" or "High Risk" alert to the doctor.

6.Web Interface(UI/UX) design:

6.1 Clinical input form: The frontend of the model will use a clean and secure form for medical professions. Categorical inputs (Sex, Chest type,...etc) will use a strict dropdown and numerical inputs(age, blood pressure,...etc) will use pre-defined medical boundariies for numerical fields. This is to prevent human errors while entering data and sending and crashing that into AI.

6.2 Diagnostic dashboard: Once the diagnosis is done, the system will give a color-coded output for readability. For 0 prediction i.e. “Healthy/Low-risk”....the system will give a green color-coded output while for prediction 1 i.e. “High Risk of Cardiovascular Disease”...the system will give the output in red.

7. Project Timeline:

Phase 1 - AI development: processed the cleveland dataset, trained the DBN and applied the Taylor-BSA optimizer.

Phase 2 - Database setup: designing the mySQL schema with two tables (`Patients_Vitals` and `AI_Predictions`) and establishing a connection between the database & python backend

Phase 3 - Designing UI/UX : Building the frontend user interface, creating the API endpoints, and securely routing the data flow from the website form, through the database, to the AI model, and back to the user's screen.